

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

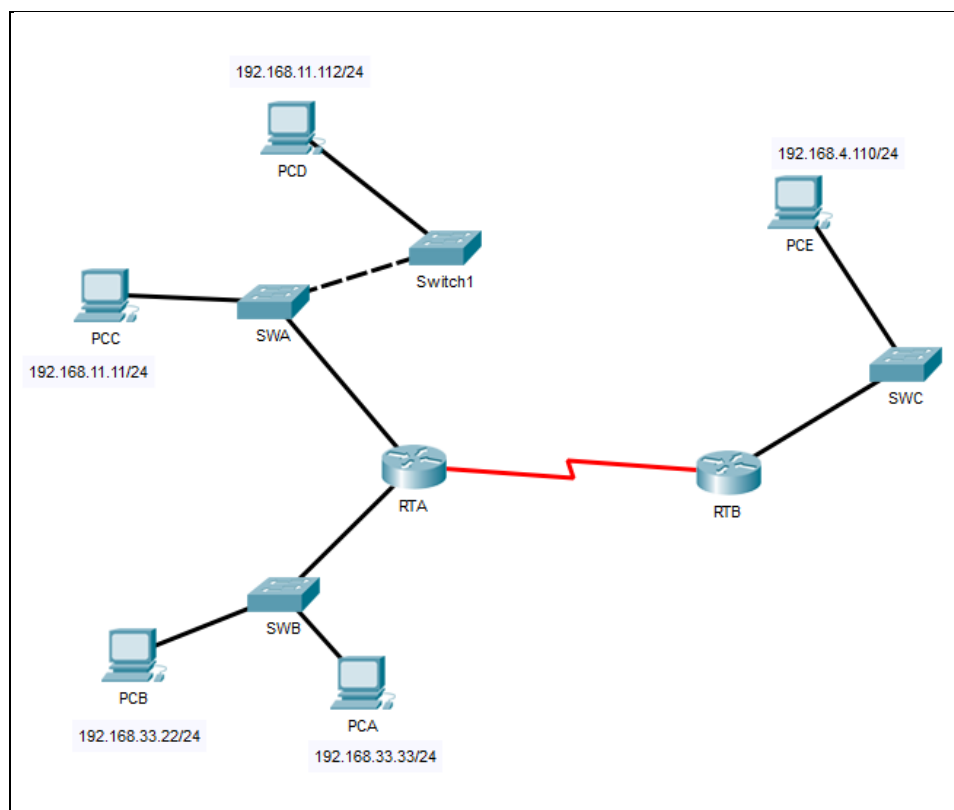


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.11.1      -         0002.4A00.0E91  ARPA   FastEthernet1/0
Internet 192.168.33.1      -         000C.CF0C.593A  ARPA   FastEthernet0/0
RTA#
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.4.1      -         0001.977A.B614  ARPA   FastEthernet0/0
RTB#
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

SWA

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0002.4a00.0e91    DYNAMIC     Fa0/1
1       000c.8546.7d85    DYNAMIC     Fa1/1
SWA#
```

SWB

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    000c.cf0c.593a    DYNAMIC Fa0/1
SWB#|
```

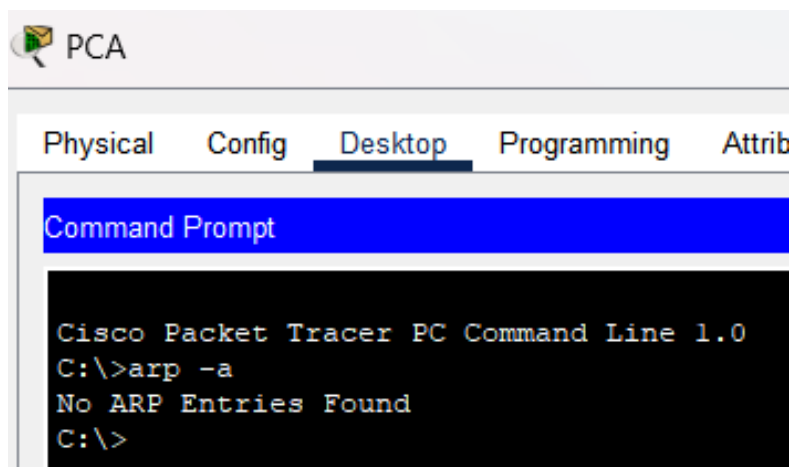
SWC

```
SWC>enable
SWC#show arp

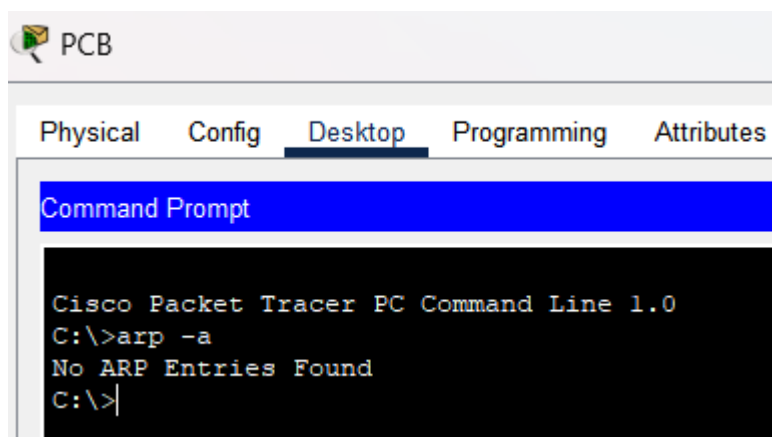
SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0001.977a.b614    DYNAMIC Fa0/1
SWC#|
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type `arp -a` and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

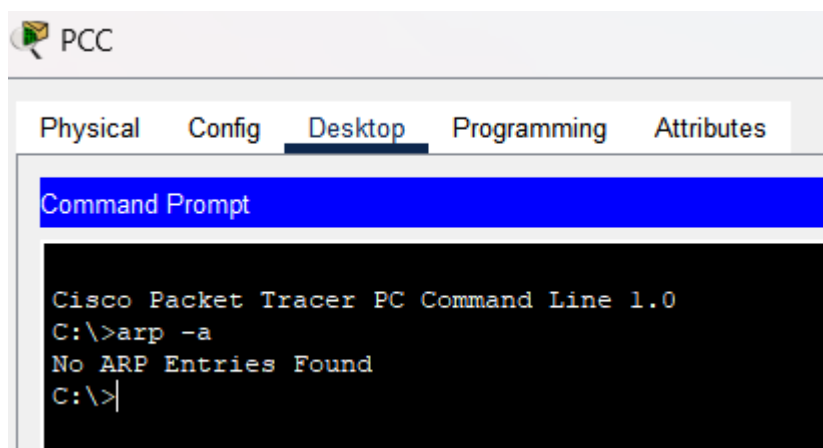
PCA



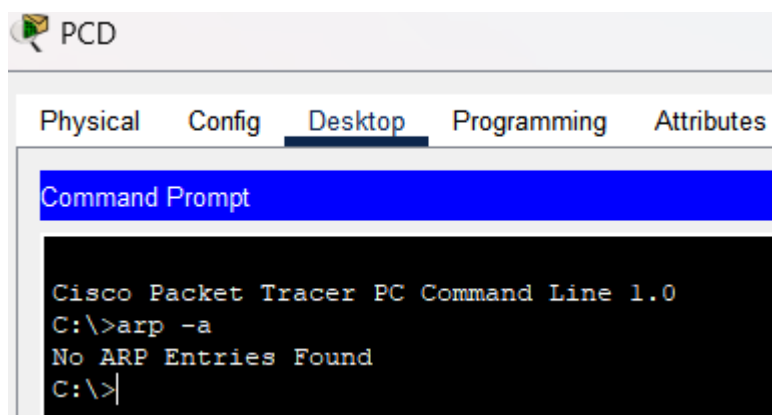
PCB

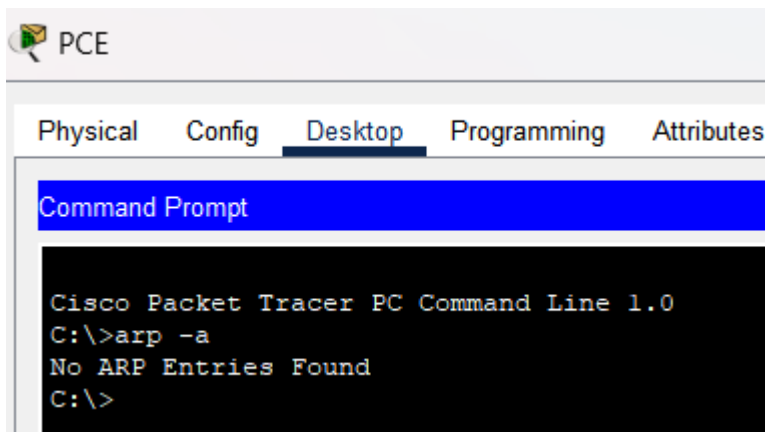


PCC



PCD





- e. What are your thoughts on the results?

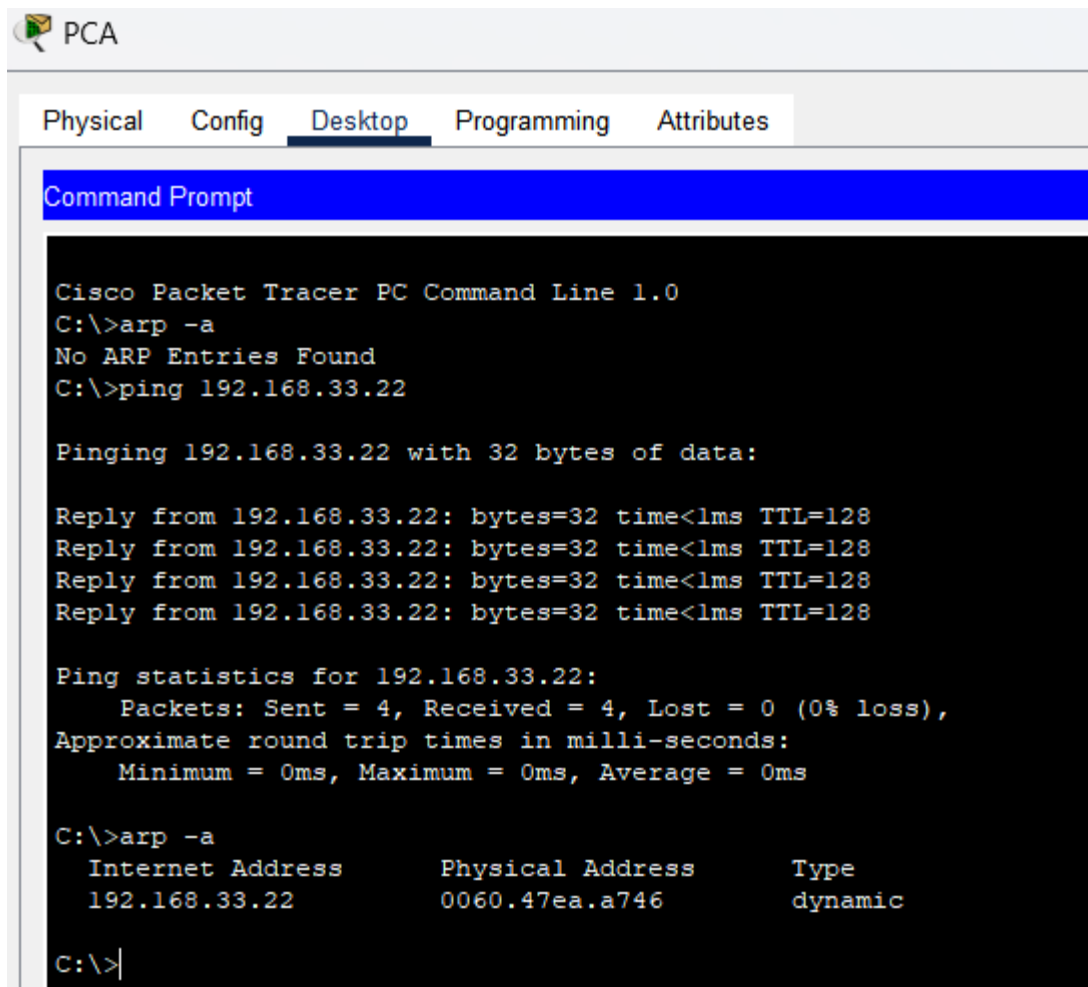
No ARP entries will be found on any PCs, as no ARP requests are sent or received, meaning the ARP table on the PCs will remain empty. Typically, ARP entries are created when a device sends an ARP request and receives a reply, mapping an IP address to a MAC address. Since this communication does not occur in this case, the ARP table will not have any entries.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.
- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.



The screenshot shows the 'PCA' window in Cisco Packet Tracer. The 'Desktop' tab is selected, and the 'Command Prompt' application is open. The command prompt displays the output of the following commands:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

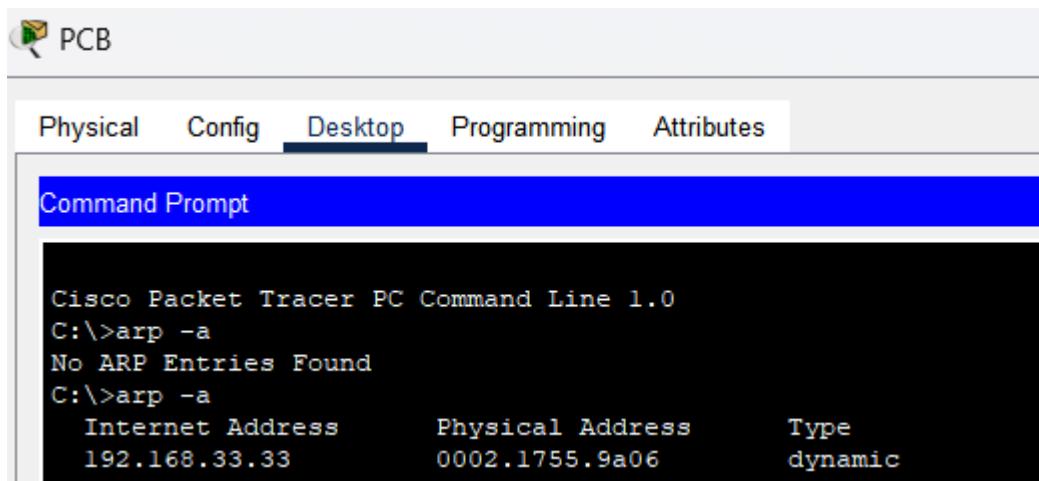
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
    Internet Address      Physical Address      Type
    192.168.33.22         0060.47ea.a746       dynamic

C:\>|
```

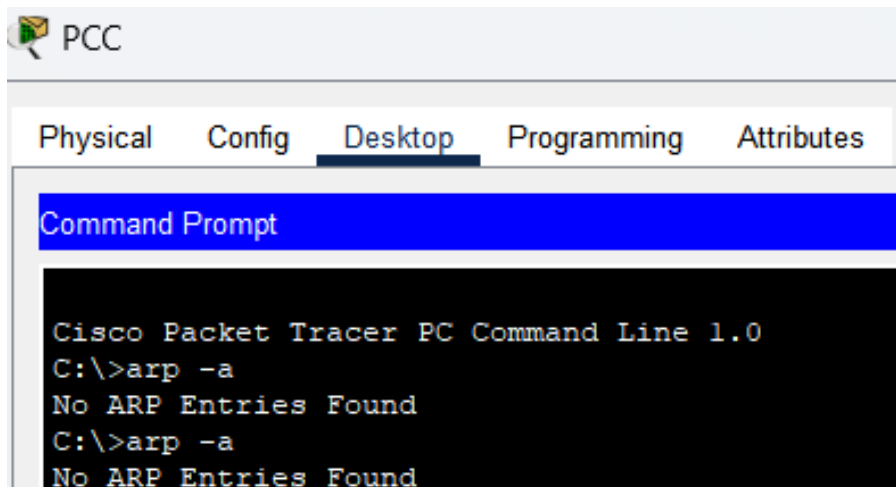
- d. In the Command prompt of PCB, type **arp -a**. Paste the result of this command here



The screenshot shows the 'PCB' window in Cisco Packet Tracer. The 'Desktop' tab is selected, and the 'Command Prompt' application is open. The command prompt displays the output of the following commands:

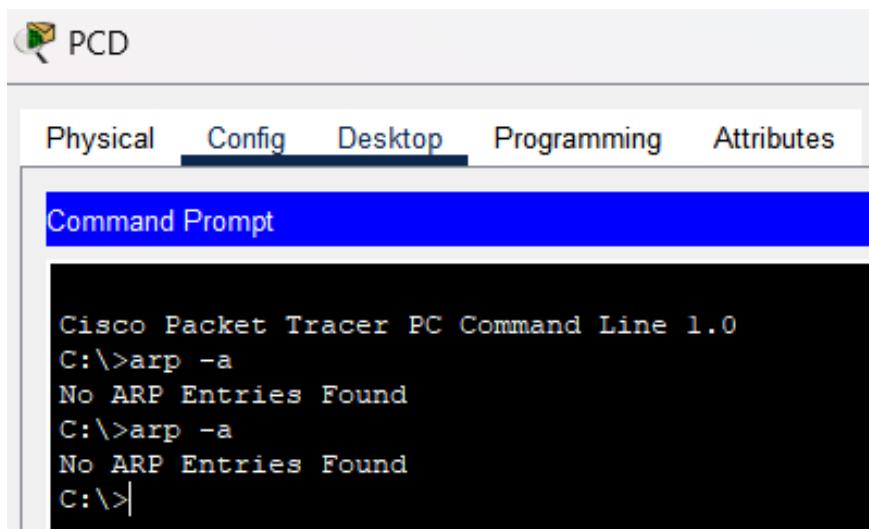
```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
    Internet Address      Physical Address      Type
    192.168.33.33         0002.1755.9a06       dynamic
```

- e. In the Command prompt of PCC, PCD and PCE, type **arp -a**. Paste the result of this command here.



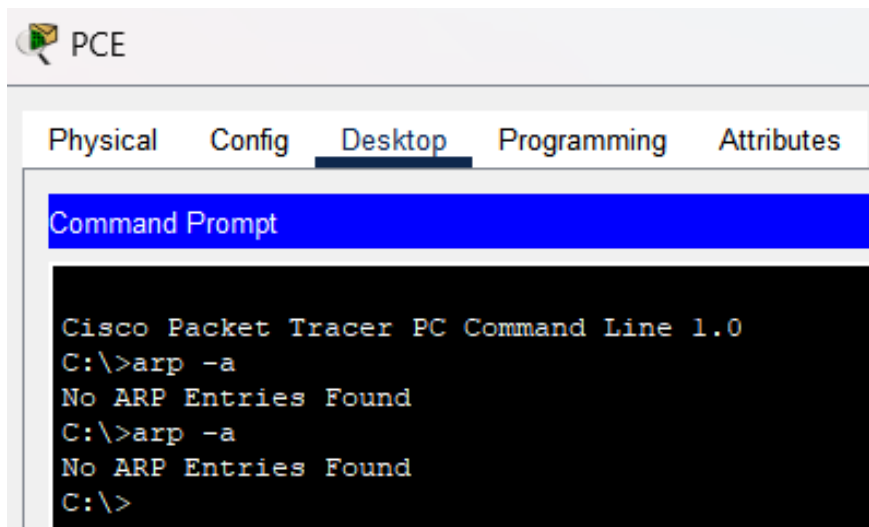
The screenshot shows the PCC (Packet Configuration Console) window with the 'Desktop' tab selected. The 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
```



The screenshot shows the PCD (Packet Configuration Console) window with the 'Desktop' tab selected. The 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>
```

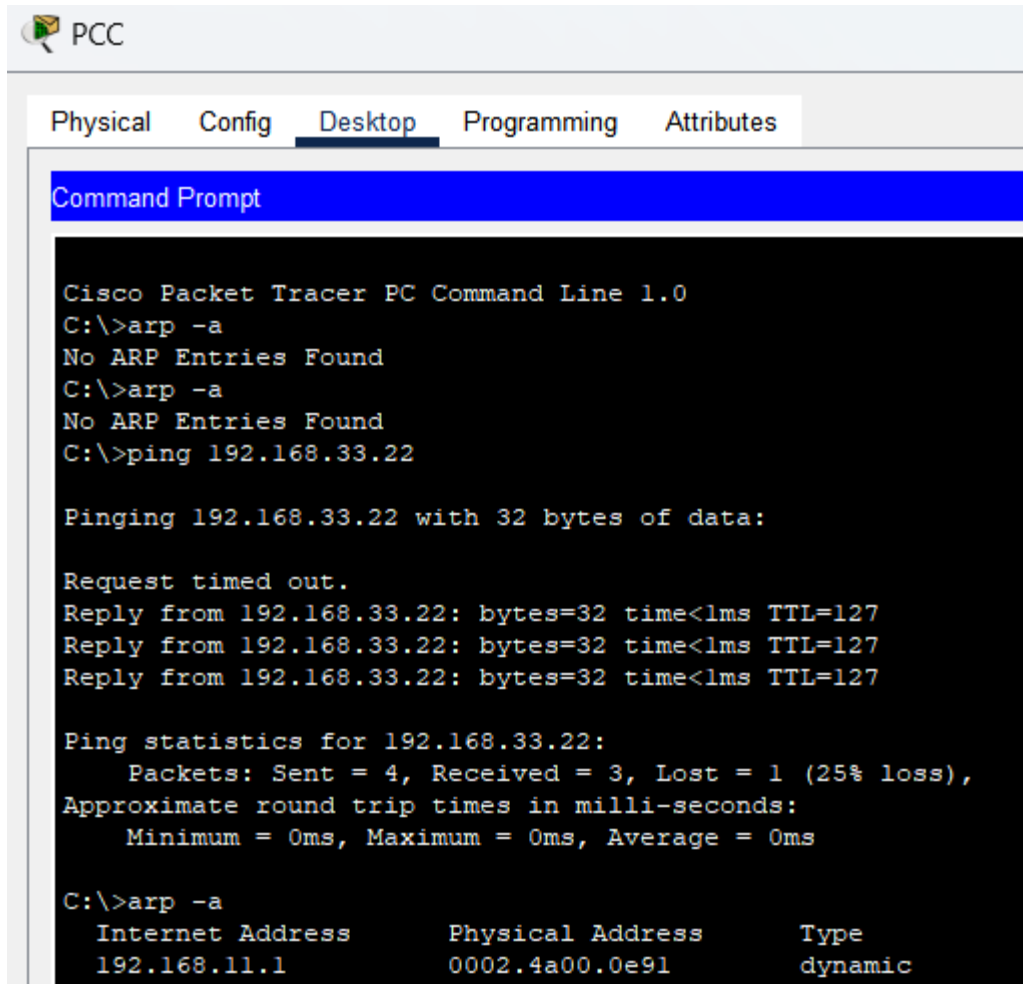


The screenshot shows the PCE (Packet Configuration Console) window with the 'Desktop' tab selected. The 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>
```

Step 2: Generate traffic between PCC to all other PC except PCA.

- a. Click **PCC** and click the Desktop tab > Command Prompt.
- b. Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.



The screenshot shows the Cisco Packet Tracer interface for PC PCC. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time<lms TTL=127
Reply from 192.168.33.22: bytes=32 time<lms TTL=127
Reply from 192.168.33.22: bytes=32 time<lms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
```

Internet Address	Physical Address	Type
192.168.11.1	0002.4a00.0e91	dynamic

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.


```

C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time=1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>arp -a
    Internet Address      Physical Address      Type
    192.168.11.1          0002.4a00.0e91       dynamic
    192.168.11.112        0001.6462.0278       dynamic

C:\>

```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```

C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=7ms TTL=126
Reply from 192.168.4.110: bytes=32 time=16ms TTL=126
Reply from 192.168.4.110: bytes=32 time=15ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 16ms, Average = 12ms

C:\>arp -a
    Internet Address      Physical Address      Type
    192.168.11.1          0002.4a00.0e91       dynamic
    192.168.11.112        0001.6462.0278       dynamic

```

- e. Discuss the results you got from all the commands on PCC.

The ARP entries are revealed in the ARP table only after PCC pings PCB, PCD, and PCE. No ARP entries are found before these ping activities. The ARP table displays entries once PCB, PCD, and PCE receive the ARP request from PCC and subsequently receive pings from PCB, PCD, and PCE.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11     5          00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22     5          0060.47EA.A746  ARPA   FastEthernet0/0
RTA#
```

- g. At Router RTB, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.4.1       -          0001.977A.B614  ARPA   FastEthernet0/0
Internet  192.168.4.110     4          0060.702D.7C08  ARPA   FastEthernet0/0
RTB#
```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
```

```
SWA#show arp
```

```
SWA#show mac-address-table
```

```
Mac Address Table
```

```
-----  
Vlan    Mac Address      Type      Ports  
----    -  
1       0001.6462.0278   DYNAMIC   Fa1/1  
1       0002.4a00.0e91   DYNAMIC   Fa0/1  
1       000c.8546.7d85   DYNAMIC   Fa1/1  
1       00d0.d39a.c0d9   DYNAMIC   Fa2/1  
SWA#
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable  
SWB#show arp  
  
SWB#show mac-address-table
```

```
SWB>enable
```

```
SWB#show arp
```

```
SWB#show mac-address-table
```

```
Mac Address Table
```

```
-----  
Vlan    Mac Address      Type      Ports  
----    -  
1       000c.cf0c.593a   DYNAMIC   Fa0/1  
SWB#
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0001.977a.b614    DYNAMIC Fa0/1
      1    0060.702d.7c08    DYNAMIC Fa1/1
```

```
Switch>enable
Switch#show arp

Switch#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0005.5e95.187b    DYNAMIC Fa0/1
Switch#
```

- d. Do switches use arp table? (Y/N)

Yes

- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation.*

It is not part of NetComm syllabi, it is just for knowledge..

Only Layer 3 switches utilize ARP tables because ARP is a Layer 3 protocol that resolves IP addresses to MAC addresses for routing between subnets. In contrast, basic Layer 2 switches use only MAC address tables at the Data-Link Layer to forward frames within the same network segment and, therefore, do not utilize ARP tables.

f. What information does the command `show mac-address-table` give?

The `show mac-address-table` command gives the list of the MAC addresses the switch has learned together with their corresponding details. It contains the MAC Address, which is the unique hardware address assigned to all devices attached to the switch; the VLAN, which describes the VLAN ID with which each MAC address is associated; the Type, which says whether the MAC address is dynamic-like, learned automatically, or static, manually configured; and the Port, which indicates the switch port that the MAC address is associated with, giving its physical connection. The term "DYNAMIC" here refers specifically to MAC addresses that the switch has learned dynamically in its operations.

Part 3: Attach wireless lab results.

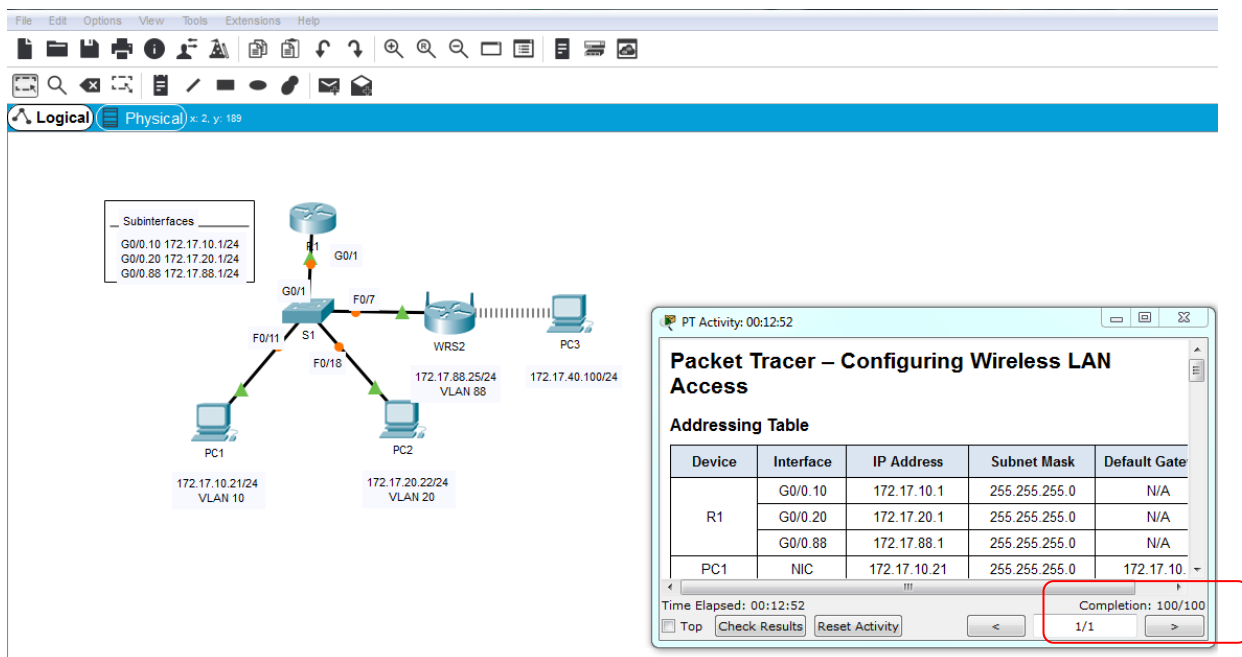
In this part, you will use Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- a. Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*

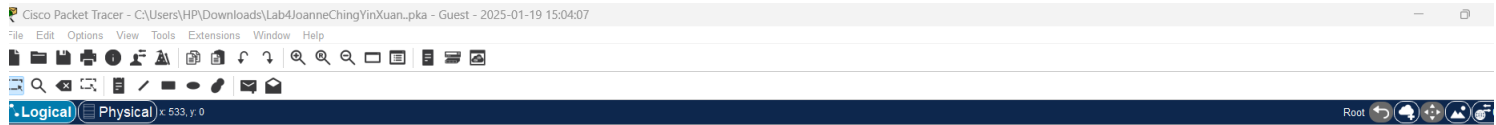
Cisco Packet Tracer - C:\Users\HP\Downloads\Lab4JoanneChingYinXuan.pka - Joanne Ching Yin Xuan - 2025-01-19 14:47:41

- b. Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.



****The step by step screenshot is behind section c**

- c. Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.



PT Activity: 00:14:04

Packet Tracer – Configuring Wireless LAN Access

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1
PC2	NIC	172.17.20.22	255.255.255.0	172.17.20.1
PC3	NIC	DHCP Assigned	DHCP Assigned	DHCP Assigned
WRS2	NIC	172.17.88.25	255.255.255.0	172.17.88.1

Objectives

- Part 1: Configure a Wireless Router
- Part 2: Configure a Wireless Client
- Part 3: Verify Connectivity

Scenario

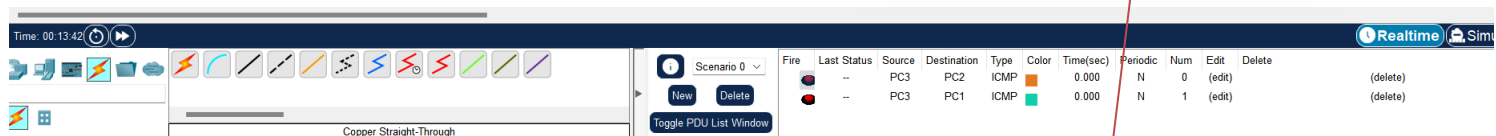
In this activity, you will configure a wireless router, allowing for remote access from PCs as well as wireless connectivity with WPA2 security. You will manually configure PC wireless connectivity by entering the wireless router SSID and password.

Part 1: Configure a Wireless Router

Time Elapsed: 00:14:04

☐ Top ☐ Dock

Completion: 100/100



PT Activity: 00:14:04

Packet Tracer – Configuring Wireless LAN Access

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1
PC2	NIC	172.17.20.22	255.255.255.0	172.17.20.1
PC3	NIC	DHCP Assigned	DHCP Assigned	DHCP Assigned
WRS2	NIC	172.17.88.25	255.255.255.0	172.17.88.1

Objectives

- Part 1: Configure a Wireless Router
- Part 2: Configure a Wireless Client
- Part 3: Verify Connectivity

Scenario

In this activity, you will configure a wireless router, allowing for remote access from PCs as well as wireless connectivity with WPA2 security. You will manually configure PC wireless connectivity by entering the wireless router SSID and password.

Part 1: Configure a Wireless Router

Time Elapsed: 00:14:04

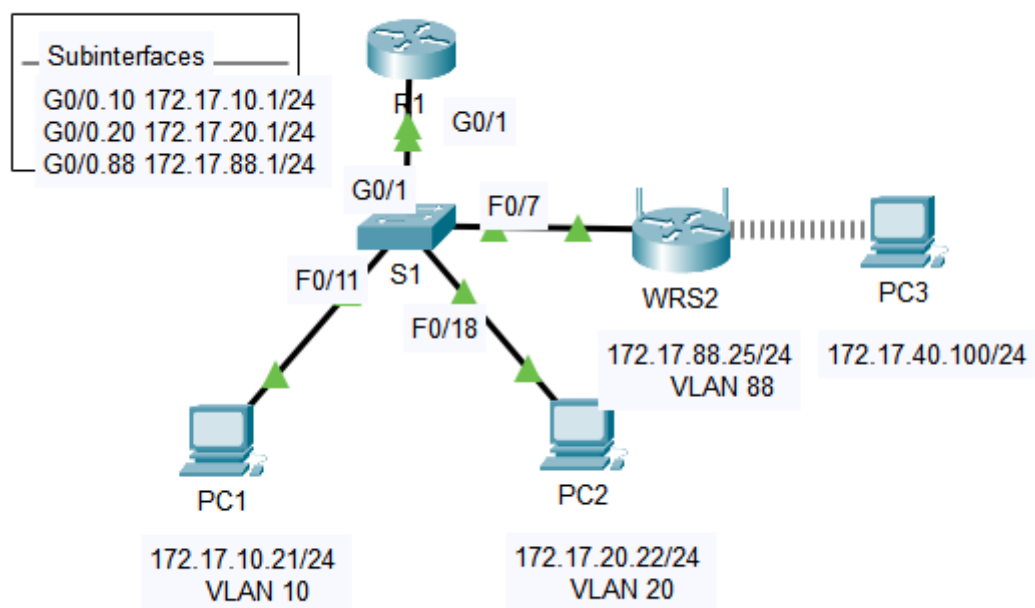
☐ Top ☐ Dock

Completion: 100/100

Part 1: Configure a Wireless Router

Step 1: Connect the Internet interface of WRS2 to S1.

Connect the **WRS2** Internet interface to the **S1** F0/7 interface.



Step 2: Configure the Internet connection type.

- Click **WRS2 > GUI** tab.
- Set the **Internet Connection type** to **Static IP**.
- Configure the IP addressing according to the Addressing Table.

Internet Setup	
Internet Connection type	Static IP ▼
	Internet IP Address: 172 . 17 . 88 . 25
	Subnet Mask: 255 . 255 . 255 . 0
	Default Gateway: 172 . 17 . 88 . 1
	DNS 1: 0 . 0 . 0 . 0
	DNS 2 (Optional): 0 . 0 . 0 . 0
	DNS 3 (Optional): 0 . 0 . 0 . 0
Optional Settings (required by some internet service providers)	Host Name:
	Domain Name:
	MTU: ▼ Size: 1500

Step 3: Configure the network setup.

- Scroll down to **Network Setup**. For the **Router IP** option, set the IP address to **172.17.40.1** and the subnet mask to **255.255.255.0**.
- Enable the DHCP server.
- Scroll to the bottom of the page and click **Save Settings**.

Network Setup

Router IP

DHCP Server Settings

IP Address:

172

17

40

1

Subnet Mask:

255.255.255.0

DHCP Server:

☒ Enabled
☐ Disabled

DHCP Reservation

Start IP Address: 192.168.1.

100

Maximum number of Users:

50

IP Address Range: 192.168.1. 100 - 149

Client Lease Time:

0

minutes (0 means one day)

Static DNS 1:

0

0

0

0

Static DNS 2:

0

0

0

0

Static DNS 3:

0

0

0

0

WINS:

0

0

0

0

Save Settings

Cancel Changes

Step 4: Configure wireless access and security.

- At the top of the window, click **Wireless**. Set the **Network Mode** to **Wireless-N Only** and change the SSID to **WRS_LAN**.
- Disable **SSID Broadcast** and click **Save Settings**.

Physical
Config
GUI
Attributes

Wireless

Setup

Wireless

Security

Access Restrictions

Applications & Gaming

Wireless-N Advanced

Basic Wireless Settings

Wireless Security

Guest Network

Wireless MAC Filter

Basic Wireless Settings

Network Mode:

Wireless-N Only

Network Name (SSID):

WRS_LAN

Radio Band:

Auto

Wide Channel:

Auto

Standard Channel:

1 - 2.412GHz

SSID Broadcast:

☐ Enabled
☒ Disabled

- c. Click the **Wireless Security** option.
- d. Change the **Security Mode** from **Disabled** to **WPA2 Personal**.
- e. Configure **cisco123** as the passphrase.
- f. Scroll to the bottom of the page and click **Save Settings**.

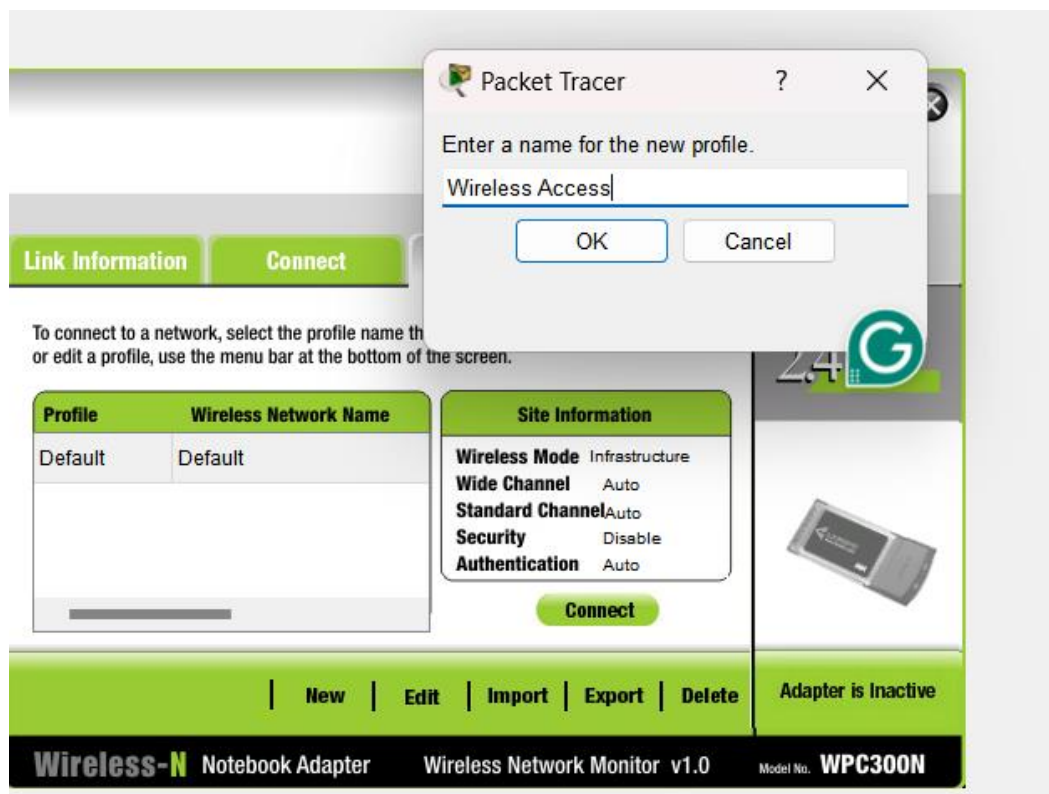
The screenshot shows the 'Wireless Security' configuration page. The top navigation bar includes 'Wireless', 'Setup', 'Wireless', 'Security', 'Access Restrictions', 'Applications & Gaming', and 'Adminis'. Below this, the 'Wireless Security' section is active, showing settings for 'Security Mode' (WPA2 Personal), 'Encryption' (AES), 'Passphrase' (cisco123), and 'Key Renewal' (3600 seconds).

Part 2: Configure a Wireless Client

Step 1: Configure PC3 for wireless connectivity.

Because SSID broadcast is disabled, you must manually configure **PC3** with the correct SSID and passphrase to establish a connection with the router.

- a. Click **PC3 > Desktop > PC Wireless**.
- b. Click the **Profiles** tab.
- c. Click **New**.
- d. Name the new profile **Wireless Access**.



- e. On the next screen, click **Advanced Setup**. Then manually enter the SSID of **WRS_LAN** on **Wireless Network Name**. Click **Next**.

Creating a Profile

Wireless Mode

Please choose the Wireless Mode that best suits your needs.

☒ **Infrastructure Mode**
Select Infrastructure Mode if you want to connect to a wireless router or access point.

☐ **Ad-Hoc Mode**
Select Ad-Hoc Mode if you want to connect to another wireless device directly without using a wireless router or access point.

Please enter the wireless network name (SSID) for your wireless network.
The wireless network name is shared by all devices in a wireless network and is case-sensitive.

Wireless Network Name

[Back](#) | [Next](#)

Wireless-N Notebook Adapter Wireless Network Monitor v1.0 Model No. **WPC300N**

- f. Choose **Obtain network settings automatically (DHCP)** as the network settings, and then click **Next**.

Creating a Profile

Network Settings

☒ **Obtain network settings automatically (DHCP)**
Select this option to have your network settings assigned automatically.

☐ **Specify network settings**
Select this option to specify the network settings for the adapter.

IP Address

DNS 1

Subnet Mask

DNS 2

Default Gateway

Back

Next

- g. On **Wireless Security**, choose **WPA2-Personal** as the method of encryption and click **Next**.

Creating a Profile

Wireless Security

Security

WPA2-Personal

Please select the wireless security method used by your existing wireless network.

WEP stands for Wired Equivalent Privacy.

WPA-Personal, also known as Pre-shared Key, is a security standard stronger than WEP encryption.

WPA2-Personal is the newer version with stronger encryption than WPA-Personal.

WPA-Enterprise, **WPA2-Enterprise** and **RADIUS** use Remote Authentication Dial-In User Service (RADIUS).

Back

Next

- h. Enter the passphrase **cisco123** and click **Next**.

Creating a Profile

Wireless Security - WPA2 Personal

Pre-shared Key: [Masked Password]

Please enter a Pre-shared Key that is 8 to 63 characters in length.

Back | **Next**

Wireless-N Notebook Adapter | Wireless Network Monitor v1.11 | Model No. **WPC300N**

- i. Click **Save** and then click **Connect to Network**.

Step 2: Verify PC3 wireless connectivity and IP addressing configuration.

- a. The **Signal Strength** and **Link Quality** indicators should show that you have a strong signal.

Link Information | **Connect** | **Profiles**

Back | **Statistics** | **Save to Profile**

Wireless Network Status

Radio Band	20MHz	Network Type	Mixed B/G/N
Wireless Network Name	WRS_LAN	IP Address	172.17.40.100
Wireless Mode	Infrastructure	Subnet Mask	255.255.255.0
Wide Channel	N/A	Default Gateway	172.17.40.1
Standard Channel	1 - 2.412GHz	DNS1	0.0.0.0
Security	WPA2-Personal	MAC Address	0050.0F3D.9806
Authentication	Auto		

2.4GHz

Signal Strength [Full Bar] | **Link Quality** [Full Bar]

Adapter is Active

Wireless-N Notebook Adapter | Wireless Network Monitor v1.0 | Model No. **WPC300N**

- b. Click **More Information** to see details of the connection including IP addressing information.

Confirm New Settings

Profile Settings

Wireless Network Name	WRS_LAN	IP Address	Auto
Wireless Mode	Infrastructure	Subnet Mask	Auto
Network Mode	Mixed Mode	Default Gateway	Auto
Radio Band	Auto	DNS1	Auto
Wide Channel	Auto	DNS2	
Standard Channel	Auto		
Security	WPA2 Personal		
Authentication	Auto		

Exit | Back | Save

Wireless-N Notebook Adapter | Wireless Network Monitor v1.0 | Model No. WPC300N

- c. Close the **PC Wireless** configuration window.

Part 3: Verify Connectivity

All the PCs should have connectivity with one another.

```
C:\>ping 172.17.10.21

Pinging 172.17.10.21 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 172.17.10.21: bytes=32 time=17ms TTL=126
Reply from 172.17.10.21: bytes=32 time=22ms TTL=126

Ping statistics for 172.17.10.21:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 22ms, Average = 19ms

C:\>ping 172.17.10.21

Pinging 172.17.10.21 with 32 bytes of data:

Reply from 172.17.10.21: bytes=32 time=29ms TTL=126
Reply from 172.17.10.21: bytes=32 time=20ms TTL=126
Reply from 172.17.10.21: bytes=32 time=22ms TTL=126
Reply from 172.17.10.21: bytes=32 time=16ms TTL=126

Ping statistics for 172.17.10.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 29ms, Average = 21ms
```

```
C:\>ping 172.17.20.22

Pinging 172.17.20.22 with 32 bytes of data:

Request timed out.
Reply from 172.17.20.22: bytes=32 time=9ms TTL=126
Reply from 172.17.20.22: bytes=32 time=18ms TTL=126
Reply from 172.17.20.22: bytes=32 time=21ms TTL=126

Ping statistics for 172.17.20.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 21ms, Average = 16ms

C:\>ping 172.17.20.22

Pinging 172.17.20.22 with 32 bytes of data:

Reply from 172.17.20.22: bytes=32 time=32ms TTL=126
Reply from 172.17.20.22: bytes=32 time=18ms TTL=126
Reply from 172.17.20.22: bytes=32 time=9ms TTL=126
Reply from 172.17.20.22: bytes=32 time=31ms TTL=126

Ping statistics for 172.17.20.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 32ms, Average = 22ms

C:\>|
```